

Haoyang Jiang

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Education

- **Ph.D. in Data Science** *Expected Graduation: 2027*
College of William & Mary, Williamsburg, VA
- **M.S. in Astrometry and Celestial Mechanics** *Sept. 2020 – Jun. 2023*
University of Science and Technology of China, Hefei, China
- **B.S. in Geophysics (Minor in Computer Science)** *Sept. 2016 – Jun. 2020*
China University of Mining and Technology, Beijing, China

Research Projects

- **Foundation Models for Forecasting and Data Assimilation:** Adapting foundation models for observation-guided long-horizon forecasting under sparse data and distribution shift.
- **Multi-Back Flow for Full-Waveform Inversion** [\[GitHub\]](#): Improving FWI reconstruction with pretrained generative priors and inference-time optimization without task-specific retraining.
- **Second-Order Optimal-Transport Flow Matching** [\[GitHub\]](#): Unifying position and velocity through a variational formulation, improving generation quality across images, videos, and scientific systems.
- **Boundary-Aware Scientific Machine Learning: FIE-NO** [\[Paper\]](#) addresses complex boundary conditions in neural operators via integral equations; **Ghost-Node GNNs** [\[GitHub\]](#) recover missing boundary information in open physical networks as an inverse problem.
- **Physics-Guided Neural Flux Prediction** [\[GitHub\]](#): Preserving directional and high-frequency flux information on directed physical networks with physics-consistent message passing.
- **Space Systems and Uncertainty Modeling:** Modeled satellite trajectories and studied orbit-error propagation under atmospheric-density uncertainty using machine-learning and numerical methods.

Publications

1. Jiang H. Y., Qu Y. Z. *Fredholm Integral Equations Neural Operator (FIE-NO) for Boundary Value Problems*. Accepted by **Machine Learning: Engineering**, 2026.
2. Jiang H. Y., Wang J. D., He Y., et al. *Topology-aware Neural Flux Prediction Guided by Physics*. In **ICML**, 2025.
3. Jiang H. Y., Zhang M. J., et al. *Orbital error propagation considering atmospheric density uncertainty*. **Advances in Space Research**, 71(6), 2566–2574, 2023.
4. Jiang H. Y., Qu Y. Z. *Transfer Operator Learning with Fusion Frame*. arXiv:2408.10458, 2024.
5. Jiang H. Y., Wang Z., Gao S., Zhang Y. J., Zhu X., He Y. *Physics-Refined Spatiotemporal Forecasting on Open-Boundary Hydrologic Graphs*. In **IEEE ICDM**, 2026.
6. Jiang H. Y., Qu B., Zhu X., Tan J., He Y. *Boundary-Consistent Graph Neural Networks for Topological Flux Prediction*. **Transactions on Machine Learning Research (TMLR)**, 2026.

Honors and Awards

- **Outstanding Graduate Student Award**, University of Science and Technology of China, 2022, 2023
- **First-Class Scholarship**, China University of Mining and Technology (Beijing), 2018, 2019

Conferences for Invited Talks

- *Topology-Aware Graph Neural Networks by Physics*. KGML Workshop: Leading the New Paradigm of AI for Science, University of Michigan, Ann Arbor, Aug. 2025.
- *Satellite trajectory modeling using machine learning methods*. International Symposium on Space Flight Dynamics (ISSFD), Beijing, Aug. 2022.
- *Orbit error propagation considering atmospheric density uncertainty*. Youth Innovation Promotion Association of the Chinese Academy of Sciences, Nanjing, Dec. 2021.

Professional Service

- **Reviewer:** MM (2025), ICDM (2025, 2026), AAAI (2026), ICLR (2026), KDD (2026), NeurIPS (2026)

Internship Experience

ORIGIN SPACE, Nanjing, China

Feb. 2022 – Jun. 2022

Assistant Engineer Intern

- Applied machine-learning and numerical methods in a space-technology R&D setting, reproducing and benchmarking research algorithms on engineering data.
- Improved research prototypes through data preprocessing, parameter tuning, and runtime optimization for engineering deployment.

Skills

- **Machine Learning:** Neural Operators, Physics-Informed ML, Graph Neural Networks, Generative Models
- **Scientific Computing:** Numerical Algorithms, PDE/ODE Solvers, Inverse Problems
- **Programming:** Python, MATLAB, and relevant ML libraries